



AMET-PHYSICS HORIZONS

Academic Year: 2024-25

ODD Semester Magazine

AMET Deemed to be University, Chennai – 603 112

EDITORIAL

Dear Readers

AMET-PHYSICS HORIZONS is the official magazine of the Physics Department at AMET University, highlighting the department's growth and faculty achievements during the odd semester. This issue covers the academic contributions, research, and professional development of our faculty members, along with their involvement in various workshops and seminars. It also celebrates the department's progress, fostering a culture of innovation and excellence in the field of Physics. Through this magazine, we reflect on our collective achievements and the department's continued success.

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President

Dr. Rajesh Ramachandran

Vice-Chancellor

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Editor-in-Chief

Dr. K. Anandan, Associate Professor and Head
Department of Physics, AMET Deemed to be University

Published By

Department of Physics & Centre for Internal Publication
Owned by Academy of Maritime Education and Training (AMET)
Deemed to be University
135, East Coast Road, Kanathur - 603 112.
Chennai, India, Phone: 044-27472155

FOREWORD



It is with great pleasure that I present this edition of the Physics Department's Academic year 2024-25 Odd Semester Magazine, a reflection of the continued growth and vibrant academic culture we foster at AMET University. As a department dedicated to preparing the next generation of leaders in technical and maritime fields, our primary goal remains to equip undergraduate students with not only a solid understanding of fundamental physics but also the practical skills and critical thinking required to excel in their respective careers. Our faculty members, who are passionate about both teaching and research, continue to make remarkable strides in a wide range of scientific domains. Through their active participation in international conferences, workshops, and FDPs, they bring the latest advancements and methodologies into our classrooms. Their involvement in research keeps our department at the forefront of innovation, and it is through these scholarly pursuits that we aim to inspire and cultivate a deeper interest in the sciences among our research students.

This semester, our department has continued to uphold its tradition of organizing events that contribute to both academic and professional development. From seminars on research methodology, intellectual property rights, skill development programs and guest lectures from industry experts, we remain committed to creating opportunities for our students to engage with emerging trends and technologies. Additionally, our involvement in submitting project proposals to government agencies ensures that our faculty members are exposed to real-world problem-solving and research applications. I am confident that the dedication, enthusiasm, and intellectual curiosity demonstrated by our faculty and students will continue to elevate the standards of our department. I am proud of the commitment to excellence demonstrated by everyone involved and look forward to the continued growth and achievements of our department. I hope that this magazine provides a comprehensive overview of the ongoing activities, research, and milestones of the Physics Department. I extend my best wishes to our students and faculty as we continue to move forward together, striving for academic excellence and innovation.

Dr. K. Anandan
Head of the Department

PREAMBLE

It gives us immense pleasure to present the magazine of Physics department, AMET University a publication that reflects the academic spirit, research excellence, and creative talents of our faculty members. This magazine highlights the department's activities, faculty's initiatives, contributions, and achievements that continue to strengthen our commitment to quality education and scientific advancement.

The Department of Physics, established in 2001–2002, plays a vital role in laying a strong foundation in science and technology. With a dedicated team of ten faculty members holding PhDs and post-doctoral experience from reputed national and international institutions, the department offers Postgraduate and Doctoral programmes designed to prepare students for careers in frontier areas of science. Our research is supported by leading national funding agencies such as DST, CSIR, and DAE, and our faculty have contributed over 300 publications in prestigious journals.

As part of **AMET University—India's first Maritime Deemed to be University** our department contributes to a unique academic environment that combines maritime focus with strong grounding in basic and applied sciences. AMET, recognized globally and accredited with **NAAC 'A' Grade**, is a hub of maritime education and innovation, serving over 3500 students across diverse faculties. The university's strong ties with global industries, international collaborations, and state-of-the-art facilities provide a vibrant platform for holistic student development. This magazine is a testament to our department's continuous efforts to promote scientific inquiry, interdisciplinary collaboration, and academic excellence. We sincerely thank all the contributors, coordinators, and editorial team members for their valuable inputs and efforts in bringing this publication to fruition.

VISION AND MISSION OF THE UNIVERSITY

VISION

To sustain identity as a World Class Leader in Maritime Education and empower learners with wholesome knowledge through progressive innovation in training, research and development which will render students a unique learning experience and a transformation impact on the Global Society.

MISSION

AMET will strive continuously to

- ❖ Impart value-based higher education and technical knowledge with uncompromising strides of an outstanding quality.
- ❖ Emerge as a Centre of Excellence inculcating skill development in recent technologies in accordance with industrial trends.
- ❖ Create World class research capabilities on par with the finest in the world and broaden student's horizons beyond classroom education.
- ❖ Nurture talent and entrepreneurship to enable all round personality development among students.
- ❖ Empower students across socio economic strata.
- ❖ Make a positive difference to society through technical education.

QUALITY POLICY

AMET is committed to provide the highest quality in education and be the most preferred institution for pursuing marine and marine related courses.

This will be achieved by consistent focus on:

- ❖ Providing a conducive, vibrant, progressive and enriching learning atmosphere.
- ❖ Teaching Excellence and Research Output.
- ❖ Global outlook and engaging with the world through learning, teaching and research.
- ❖ Providing competitive advantage in gaining employment for further academic opportunities.
- ❖ Maintaining excellent links with commerce and industry both national and international.
- ❖ Complying with all applicable requirements and continually improving the effectiveness of Quality Management System.

VISION AND MISSION OF THE DEPARTMENT

VISION

Our primary responsibility is to prepare undergraduate students for careers in their respective technical fields. In light of the fact that Physics is a bridge between technology and society, the Department of Physics is also committed to bringing an understanding of physics in particular and science in general to both the general student population and the campus community at large.

MISSION

The Department of Physics provides high quality Physics education; producing well prepared graduates. The department of Physics promotes research and creative activities of faculty members and students.

INDEX

S.No.	Content	Page No.
1	List of Faculty Members	7
2	Courses Offered	8
3	Awards and Recognitions	9
4	Professional Body Membership	13
5	E-Contents – Video Lectures	14
6	Patents	15
7	Books/Book Chapter Published	16
8	Research Article Published	18
9	Faculty Development Programs - Attended	20
10	National and International Conference - Attended	21
11	Event Organized – Guest Lectures	22
12	Event Organized – Faculty Development Program	24
13	Event Organized – Seminar on Skill Development	25
14	Event Organized – Seminar on Research Methodology	25
15	Event Organized – Workshop on UV-Vis	26
16	Event Organized – Onam Celebration – Onam 2K24	27

LIST OF FACULTY MEMBERS



Dr. Anita R Warriar.
Associate Professor & Dean

Dr. K. Anandan
Associate Professor & Head



Dr. A. Mohamed Haroon Basha
Associate Professor

Dr. K. Rajesh
Associate Professor



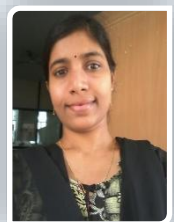
Dr. K. Gayathri
Associate Professor

Dr. Anitha Rexalin Devaraj
Associate Professor



Dr. C. Amirthakumar
Assistant Professor

Dr. Swapna Mary Gali
Assistant Professor



Mr. M. Kamalakkannan
Lab Instructor

COURSE OFFERED – ODD SEMESTER

S. No.	Course	Program
1	Nautical Physics and Electronics – I	B.Sc., Nautical Science
2	Nautical Physics and Electronics Lab – I	B.Sc., Nautical Science
3	Material Physics	B.E., Marine Engineering
4	Materials Physics Lab	B.E., Marine Engineering
5	Engineering Physics	B.E., - NAOE & ECE
6	Engineering Physics Lab	B.E., - NAOE & ECE
7	Applied Physics	B.Sc.,- RAI and DS
8	Applied Physics-I	B.Sc.,- Aviation
9	Applied Physics Lab	B.Sc., Aviation
10	Astrophysics	B.Sc., and B.E (VAC)
11	Physics for Competitive Exam	B.Sc., and B.E (VAC)

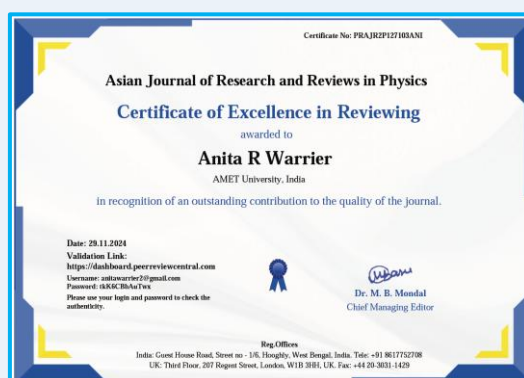
AWARDS AND RECOGNITIONS

Dr. Anita R Warriar - Professor



Dr. Anita R. Warriar Honoured with Young Scientist Award

- i) We are proud to share that **Dr. Anita R. Warriar**, *Professor and Dean (Science and Humanities), Department of Physics, AMET University*, has been conferred with the **Young Scientist Award** by the **Instrument Society of India**. This prestigious recognition was presented to her during the **National Conference on Instrumentation (NCI 2025)**, held on **January 10, 2025**, at the **Indian Institute of Science (IISc), Bangalore**. This award is a testament to Dr. Anita R Warriar's outstanding contributions in the field of instrumentation and applied physics. Her dedication to academic excellence, research innovation, and continuous mentoring of young minds has brought immense pride to the Department of Physics and to AMET University as a whole. We extend our heartfelt congratulations to Dr. Anita R. Warriar for this well-deserved recognition and wish her continued success in all future endeavors.



Certificate of Excellence in Reviewing

- ii) Dr. Anita R. Warriar has been awarded the **Certificate of Excellence in Reviewing** by the *Asian Journal of Research and Reviews in Physics*, in recognition of her outstanding contribution to maintaining and enhancing the quality of the journal. This acknowledgment reflects her commitment to academic integrity, critical analysis, and the advancement of scientific research through rigorous peer review. We commend her dedication to the scholarly community and her continued efforts in upholding the standards of scientific publishing.

Dr. Anitha Rexalin Devaraj – Associate Professor



Certificate of Completion from NPTEL

- i) Dr. Anitha Rexalin Devaraj has successfully completed the **transcription editing task** in the *Machine Learning Tool – Chitraklekha* for the NPTEL courses “**Computational Fluid Dynamics (103106119)**” and “**Geosynthetics and Reinforced Soil Structures (105106052)**”, offered by IIT Madras. This contribution reflects her active involvement in enhancing the accessibility and quality of online learning content.



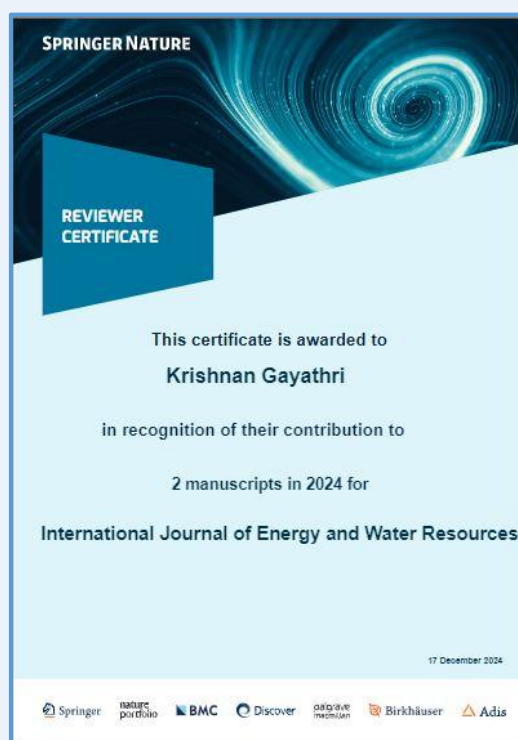
- ii) Dr. Anitha Rexalin Devaraj has been awarded the **Reviewer Certificate** in recognition of her valuable contributions to the peer review process. She has actively reviewed manuscripts for the journals *Journal of Pharmaceutical Research International*, *BP International*, *Journal of Scientific Research and Reports*, *Asian Journal of Research in Biochemistry*, and *American Journal of Quantum Chemistry and Molecular Spectroscopy*. Her dedication reflects a strong commitment to upholding the quality and integrity of scientific publications.

Dr. K. Rajesh – Associate Professor



Certificate of Reviewing

Dr. K. Rajesh received a **Certificate of Reviewing** from the journals *Optik* and *Spectrochimica Acta Part A: Molecular and Biomolecular Spectroscopy*, in recognition of his valuable contribution to the peer-review process and support in upholding the quality of scientific publications.



Reviewer Recognition by Springer Nature

Dr. K. Gayathri has been awarded the **Reviewer Certificate** by Springer Nature in recognition of her valuable contributions to the peer review process. She has actively reviewed manuscripts for the journal ***International Journal of Energy and Water Resources***. Her efforts reflect a strong commitment to maintaining the quality and integrity of scientific publications.

Dr. K. Anandan – Associate Professor and Head



Reviewer Recognition by Springer Nature

Dr. K. Anandan has received the **Reviewer Certificate** from *Springer Nature* in recognition of his valuable review contributions to the journals **Discover Applied Sciences**, **Ionics**, **Journal of Inorganic and Organometallic Polymers and Materials**, and **Journal of Sol-Gel Science and Technology**. His efforts reflect a strong commitment to maintaining the quality and integrity of scientific publications.

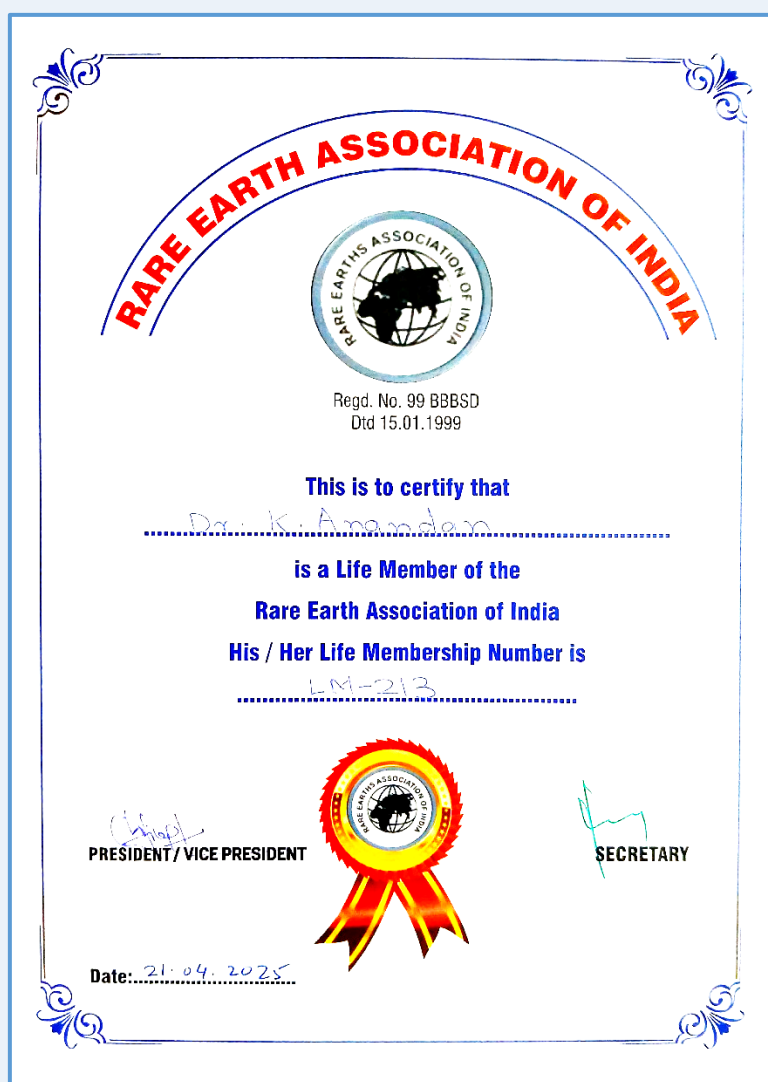
Dr. A. Mohamed Haroon Basha – Associate Professor



Faculty Achievement – Tamil Article Publication

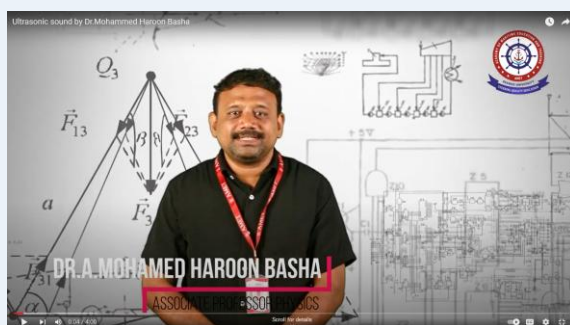
Dr. A. Mohamed Haroon Basha, Associate Professor, Department of Physics, published a Tamil article titled "மண் வளமே மனித வளம்" (*Man Valame Manitha Valam*) in **Dinamalar** on **25th August 2024**. The article highlights the significance of soil fertility as the foundation for human prosperity, reflecting his commitment to environmental awareness through regional language outreach.

Professional Body Membership



All faculty members of the Department of Physics have joined the ***Rare Earth Association of India*** as Life Members. Displayed here is a sample copy of the Life Membership Certificate issued by the Rare Earth Association of India.

Video Lectures



This series of video lectures provides in-depth knowledge on essential topics in Physics. Each lecture is delivered by a faculty member from the Department of Physics, covering fundamental concepts and real-world applications to enhance learning.

- ❖ **Dr. K. Anandan – Transformer**
Video lecture on the fundamentals and concepts of Transformers.
- ❖ **Dr. K. Rajesh – Semiconductors**
Video lecture explaining the basics and applications of Semiconductors.
- ❖ **Dr. A. Mohamed Haroon Basha – Sound**
Video lecture covering the essential principles of Sound.
- ❖ **Dr. Anitha Rexalin Devaraj – Super Conductors**
Video lecture detailing the properties and significance of Super Conductors.
- ❖ **Dr. C. Amirthakumar – Laser**
Video lecture discussing the working and applications of Lasers.

Patents



Celebrating Innovation and Excellence at AMET University

- i) We are proud to announce that **Dr. Anita R. Warriar** has achieved significant milestones in research and innovation with the award of two prestigious patents. Her ground-breaking work in the field of sustainable energy and environmental remediation has earned her recognition in the scientific community.

"Method of Producing Biodiesel from Waste Cooking Oil using MgO-SnO₂ Nanohybrids" (Patent No. 498375)

"Reusable Chalcogenide Semiconductor Nanostructures for Industrial Wastewater Purification" (Patent No. 548105)

These patents reflect Dr. Anita R Warriar's dedication to addressing pressing global challenges through innovative research. The recognition of these patents underscores the commitment of **Academy of Maritime Education and Training (AMET) Deemed to be University** in fostering cutting-edge research that contributes to the advancement of sustainable technologies and industrial solutions.

We extend our heartfelt congratulations to Dr. Anita R. Warriar for these remarkable achievements, which highlight her continuous pursuit of excellence in science and technology.

- ii) We are delighted to share that **Dr. K. Gayathri** has successfully published a patent titled *"Lightweight and Durable Polymer Composites Created through AI-Driven Material Science Research."*

Application Number: 202441085325 A

This innovative work showcases the integration of artificial intelligence with material science to develop next-generation polymer composites with enhanced strength, durability, and lightweight properties.

Book/Book Chapters Published



Outstanding Academic Contributions in Sustainable Energy and Environmental Research

We are proud to highlight the remarkable academic achievements of **Dr. Anita R. Warrier**, whose contributions in the fields of sustainable energy, green technology, and environmental remediation continue to bring distinction to **Academy of Maritime Education and Training (AMET) Deemed to be University**. Dr. Anita R. Warrier has authored significant book chapters in prestigious Springer publications:

- ❖ **Book Title:** *Circular Bioeconomy Perspectives in Sustainable Bioenergy Production*

Publisher: Springer

Chapter Title: *Green Catalysts Synthesized from Biomass for Biodiesel Production*

- ❖ **Book Title:** *Application of Microbial Technology in Wastewater Treatment and Bioenergy Recovery*

Publisher: Springer

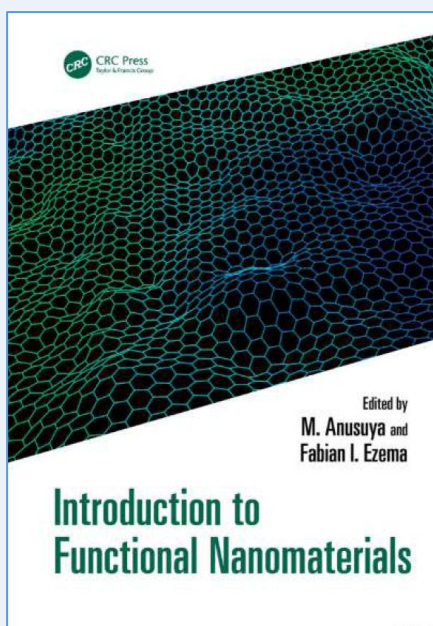
Chapter Title: *Visible-Light-Driven Photocatalysis Using Semiconductor and Plasmonic Nanoparticles for Industrial Wastewater Treatment*

In addition to her chapter contributions, Dr. Warrier has also served as an editor for a major publication:

- ❖ **Edited Book:** *Innovations in Perovskite Solar Cell Materials and Devices: Cutting-Edge Research and Practical Applications*

Publisher: IntechOpen

Her dedicated research efforts and scholarly works continue to inspire innovation and excellence in sustainable technology and advanced materials.



10

Determination of Kinetic Parameters and Mechanical Properties of Nonlinear Optical Single Crystals

P. Krishnan
St. Joseph's College of Engineering, Chennai, Tamil Nadu, India
K. Rajesh
AMET University, Chennai, Tamil Nadu, India
S. Sudhakar
Alagappa University, Karaikal, Tamilnadu, India

16.1 Introduction

The advancement in science and technology realises the importance of solo crystals for the growth of new-generation devices. The demand for high-quality crystals focuses on the importance of the rapid growth process and the extremely competent nonlinear optical crystals for technological processes. In recent years, there have been rigorous efforts to explore and develop second-order nonlinear optical materials with a high degree of efficiency for various applications such as electro-optic swapping or modulation for telecommunication and frequency conversions for laser sources at desired wavelengths.

16.1.1 Purpose of Characterisation

Characterisation renders a basis for realising and enhancing the characteristics of materials towards their applications and also comprises an appraisal of the chemical structure, structure, imperfections, thermal stability, mechanical strength, conductivity, and optical properties. If the anisotropic heat release of the crystal is large, the defined heat is low, and the thermal conductivity is low, the crystal can easily crack during the growth process or when the temperature gradient is too large. Therefore, thermal properties can be considered as important parameters for evaluating the practical viability of a particular crystal.

Hardness properties depend on the material's crystal structure and bond strength. Microhardness research was performed to apprehend the plasticity of crystals. Checking out the hardness is generally done to decide the mechanical electricity of materials. It correlates with different mechanical residences which include elastic constants and yield electricity. Hardness of the material is more suffering from dimension situations than by means of cloth residences. In this chapter, we discuss the determination of kinetic parameters and mechanical properties of nonlinear optical single crystals.

16.1.2 Thermal Analysis

Thermal analysis is a generic term that encompasses several techniques for measuring the temperature dependence of physical property parameters of matter. Thermogravimetric analysis (TGA), differential thermal analysis (DTA), and differential scanning calorimetry (DSC) are widely used techniques.

TGA involves monitoring the weight while varying temperature. The programme's turning point corresponds to the differential thermogram's peak point. Thermal properties such as thermal expansion, specific heat, and thermal conductivity of crystals greatly impact crystal growth and applications. Thermal expansion measurements also helped investigate the problem of crystal cracking. Temperature gradients

166

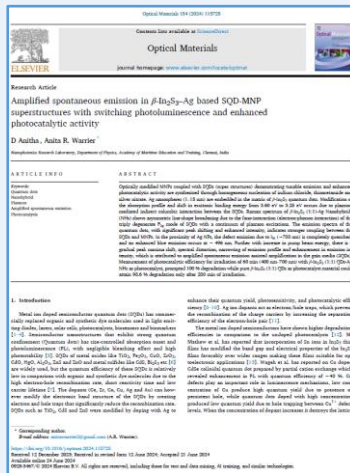
DOI: 10.1201/9781003405437-16

Academic Contribution in Advanced Materials Research

We are proud to announce that **Dr. K. Rajesh** has successfully published a book chapter in the prestigious **Taylor & Francis** publication. The book, titled *"Introduction to Functional Nanomaterials,"* features his chapter on *"Determination of Kinetic Parameters and Mechanical Properties of Nonlinear Optical Single Crystal."* This contribution highlights Dr. Rajesh's expertise in the field of nanomaterials and nonlinear optical materials, further enriching the academic and research excellence of **Academy of Maritime Education and Training (AMET) Deemed to be University**. We extend our heartfelt congratulations to Dr. K. Rajesh on this remarkable achievement and wish him continued success in his scholarly pursuits.

Research Article Publications

Our department faculty members are actively engaged in cutting-edge research across a wide range of scientific fields. As a result of their dedication and scholarly efforts, they have published numerous research articles in reputed international journals. These publications not only reflect the high academic standards of our department but also contribute significantly to the advancement of knowledge in their respective areas.



Dr. Anita R Warriar – Professor and Dean



Dr. K. Anandan – Associate Professor and Head

INVESTIGATION OF THE MOLECULAR STRUCTURE AND DFT INSIGHTS INTO (4-(4-NITROPHENYL)-9-PHENYLSULFONYL)- 9H-CARBAZOLE-2,3-DIYL)BIS(p-TOLYLMETHANONE)

M. V. Muppala¹, K. Rajesh^{2*}, D. Anitha Resaiah³,
K. Ananda⁴, K. Gayathri⁵, A. Mani⁶,
P. Devendran⁷, V. Tharyanitha⁸, P. Karinjathana⁹,
and M. Suresh Kumar⁹

Carboxylate molecules are distinguished as potential compounds in the largely emergent chemical world of heterocyclic compounds showing alike pharmacological characteristics. The knowledge of a mixture of synthetic pathways and different physicochemical parameters of such compounds are of special interest to medicinal chemists to produce combinatorial collections and carry out in-depth searches of lead molecules. Among various groups of the compounds studied, carboxylate moieties stand out as unique in features due to their biological, pharmacological, and medicinal properties. The synthesis, crystal structure, conformation, and density functional theory study of the (4-(4-nitrophenyl)-9-phenylsulfonyl)-9H-carbazole-2,3-diyl)bis(p-tolylmethanone) derivative are reported in detail. The carboxylate moiety is in the planar conformation with C-H...O hydrogen bonds and Van der Waals forces. Density functional theory is utilized to investigate the carboxylate derivative. Similar to co-crystal ligands, carboxylate compounds exhibit potent interactions with the target protein. The supramolecular properties are measured and validated using Hirshfeld surface studies.

DOI: 10.1134/S0022476624050123

Keywords: carboxylate, molecular modeling studies, Hirshfeld surface analysis, density functional theory.

INTRODUCTION

Carbazole is a heterocyclic molecule made up of nitrogen atoms. Its tricyclic structure consists of two six-membered benzene rings bonded to each other on either side by a five-membered nitrogen ring. Carbazole and its derivatives, which

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0022-4766/24/0505-0987 © 2024 by Pleiades Publishing, Ltd.

987

Nonenzymatic Selective Electrochemical Determination of Dopamine Using Fe³⁺-Doped Hydroxyapatite

Conference paper | First Online: 27 December 2024
pp 541–553 | Cite this conference paper

Proceedings of the 1st
International Conference on
Recent Advancements in...
(ICRAMST 2024)

Kurinjathana Pannesarvam, Anita R. Warrier, Roselin Rajitha Mathias, Vijayaraj Radha & Rajesh Krishnan

Abstract

The use of nanomaterials in the detection of dopamine (DA) is of significant importance in the timely identification of hyperlocomotion and Parkinson's disease. Among several scientific methodologies, electrochemical sensing has shown its efficacy as a precise, cost-effective, expeditious, and straightforward technology that enables the detection of dopamine at trace levels. Hydroxyapatite (HAp) is an inorganic compound composed of calcium phosphate, which closely resembles the mineral composition found in bones and teeth. Its great biocompatibility, outstanding electron transfer behavior, and many surface active sites which assist chemical reactions make it a prominent candidate for future developments. In this study, we present Fe³⁺-HAp ceramic nanoarchitectures that are designed to effectively detect dopamine. The synthesized samples were investigated by X-ray diffraction (XRD), FT-Raman (Raman), FT-Infrared (FTIR), Brunauer–Emmett–Teller (BET), and X-Ray Photoelectron Spectroscopy (XPS) techniques. The electrocatalytic activity of Fe³⁺-HAp-modified glassy carbon electrodes was shown to be very effective in the oxidation of DA. The oxidation peak currents exhibited a linear relationship with the concentration of DA within the range of 5–500 μM, under optimum circumstances. Linear sweep voltammetry (LSV) method was

Dr. K. Rajesh – Associate Professor



African Journal of Biological Sciences

Synthesis Of Imidazole Derivatives Using Multiple Synthetic Routes And Their Catalytic Response

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Abstract:
Synthesis of water-soluble imidazole derivatives which is attached with base sensitive fluorophore peripheral units with different organic compounds under conventional and silica supported Muffle furnace methods. Solvent free synthesis approach has more merits than the conventional method, such as non-toxic, shorter reaction time, easy workup procedure without purification process and higher yields. Synthesized substituted imidazole derivatives are acted as excellent catalyst response for the preparation of Gem-bisamides derivative under conventional approach.
Keywords: Imidazole, conventional, solid phase, optimization, Different solvent.

INTRODUCTION

Recently, carbon materials are probably the best alternative for the generation of various organic compound synthesis. Because, they have extraordinary properties, for example, huge explicit surface area, high permeable structure and solid connections among carbon and hydrogen atoms. Thus, graphene oxide is a promising contender for a wide assortment of synergistic applications. Nitrogen-containing aromatic heterocyclic compounds, particularly imidazoles, have garnered significant attention in research and industrial chemistry in recent years, mainly due to their versatile range of biological and pharmacological activities [1]. They play a pivotal role in the synthesis of biologically active molecules [2]. They play a pivotal role in the synthesis of biologically active molecules and process [3], such as anticancer, anti-aging, anticoagulant, anti-inflammatory, antimicrobial, anti-tubercular, antidiabetic, antimalarial, antiviral drugs, and enzyme inhibitors [4,5,6]. They also act as selective plant growth regulators, fungicides, herbicides, and therapeutic agents [7]. Nowadays, green chemistry and organocatalytic catalysis have extended the application of imidazoles as ionic liquids and N-heterocyclic carbenes (NHCs) [8,9]. Therefore, imidazole derivatives have become more popular due to the demand for environmentally friendly methods in chemical organic synthesis. There are several approaches for the synthesis of substituted imidazoles by condensation [10], ring cyclization [11], oxidation

Studies on the structural, optical, mechanical, and electrical properties of a novel 2-aminoanilinium benzene-1, 2-diaminium tris (4-methylbenzene-1-sulfonate) single crystal

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Received: 25 December 2023
Accepted: 10 April 2024

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Science+Business Media, LLC,
part of Springer Nature, 2024

ABSTRACT

2-Aminoanilinium benzene-1, 2-diaminium tris (4-methylbenzene-1-sulfonate) organic single crystal was grown by slow evaporation solution growth technique at ambient temperature. The grown crystal was subjected to single-crystal X-ray diffraction studies to confirm the crystal structure. The presence of different molecular groups and modes of vibrations was identified by FTIR analysis. UV-visible spectral studies revealed the transparency, lower cut-off wavelength, bandgap energy, optical parameters such as refractive index (*n*), extinction coefficient (*k*), optical conductivity, and electrical conductivity. Laser-induced damage threshold was calculated as 3.5 GW/cm². Mechanical properties of grown crystal were investigated by Vicker's micro-hardness test. The dielectric measurement at room temperature showed the electrical behavior of the crystal.

1 Introduction

The nonlinear optical materials have an appreciable attention due to their applications in various optical devices, like that opto-electronic device, frequency conversion, optical communications, optical computing, optical disk data storage device, THz communication, THz imaging, as well as military technology, etc. [1]. Nonlinearity of optical materials is due to the presence of delocalized π -electron systems connecting the donor and acceptor groups. In nonlinear optical materials, the influence of an electric field causes

distortion in the spatial distribution of electrons and nucleus of the material, and this distortion of electric dipoles creates or results in polarization. This induced polarization is multiplying the fundamental frequency to second and third harmonics. The third-order nonlinear optical phenomena in organic crystals received the importance due to their potential applications of optical switching devices, optical limiters, and optical gates. The third-order nonlinear optical parameters such as third-order NLO refractive index (*n*₃) and third-order nonlinear susceptibility (χ ₍₃₎) of 2-Aminoanilinium benzene-1, 2-diaminium tris

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<https://doi.org/10.1007/s0054-024-12584-7>

Published online: 03 May 2024



Dr. C. Amirthakumar – Assistant Professor

Faculty Development Program

Our department faculty members actively participate in Faculty Development Programs (FDPs) to continuously enhance their teaching skills, stay updated with the latest advancements in pedagogy, and promote excellence in the learning process. These programs enable our faculty to integrate innovative teaching methodologies and modern educational tools, contributing to the overall academic growth of both educators and students.



National and International Conferences Attended

Our department faculty members actively participate in various international and national conferences to share their research findings, engage with the global academic community, and stay informed about the latest developments in their fields. By presenting their research outcomes at these prestigious platforms, our faculty not only contribute to the broader scientific dialogue but also gain valuable insights that further enrich their teaching and research practices.



Event Organized – Guest Lecture

Our Physics Department actively organizes a variety of academic events such as Faculty Development Programs (FDPs), Guest Lectures, Research Methodology Workshops, and other scholarly activities. These initiatives aim to foster a vibrant academic environment by facilitating knowledge exchange, enhancing research skills, and promoting continuous professional development. By conducting these events, the department not only strengthens the academic community but also encourages collaboration, innovation, and excellence among faculty members and students.



Guest Lecture on "Semiconductor Devices and Applications"

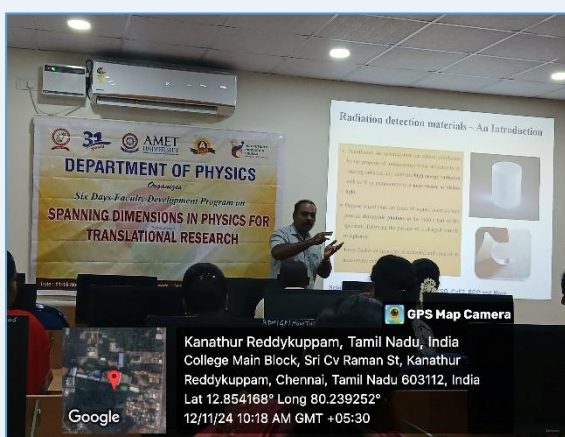
A guest lecture on "*Semiconductor Devices and Applications*" was conducted on 7th November 2024. The session was delivered by Dr. P. Krishnan, Associate Professor, St. Joseph's College of Engineering (Autonomous), Chennai. The lecture provided valuable insights into the fundamentals and advanced applications of semiconductor devices, enriching the knowledge of both students and faculty members.



Guest Lecture on "Next-Generation MEMS Oscillators: Innovations and Advancements Beyond Quartz"

The Department of Physics, in collaboration with AMET-Institution's Innovation Council (AMET-IIC), organized a guest lecture on *"Next-Generation MEMS Oscillators: Innovations and Advancements Beyond Quartz"* on 11th November 2024. The session was delivered by Dr. S. Suresh, Professor and Research Advisor, Department of Physics, Saveetha Engineering College (Autonomous), Chennai. The expert highlighted the latest innovations in Micro-Electro-Mechanical Systems (MEMS) technology, emphasizing the transition from traditional quartz-based oscillators to advanced MEMS-based systems. The lecture provided participants with valuable insights into design principles, performance enhancements, and real-world applications of MEMS oscillators in modern electronics. The event sparked engaging discussions and was particularly enriching for students and faculty members interested in sensor technology and emerging electronic components.

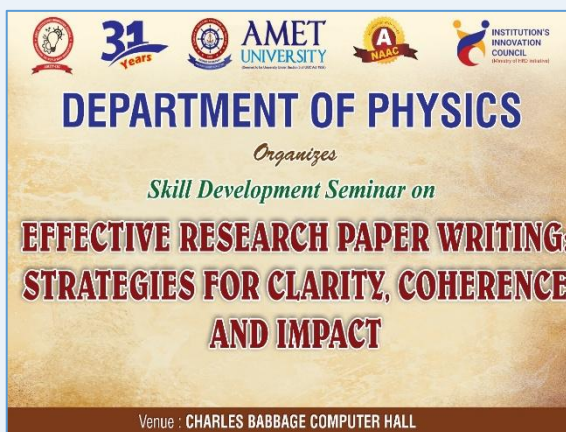
Event Organized – Faculty Development Program



FDP on "Spanning Dimensions in Physics for Translational Research"

The Six-Day Faculty Development Program (FDP) on "*Spanning Dimensions in Physics for Translational Research*", organized by the Department of Physics from 11th to 16th November 2024, brought together a range of expert speakers from various domains of physics. The sessions covered diverse and emerging areas, offering valuable insights into cutting-edge research and practical applications. The program was highly beneficial to faculty members, enriching their academic knowledge and enhancing their capacity for interdisciplinary research. The event was convened by Dr. K. Gayathri, Associate Professor and Deputy Head of the Department, and was well-received by all participants.

Event Organized – Seminar on Skill Development Program



Seminar on Skill Development on Research Paper Writing

The Department of Physics organized a Skill Development Seminar on *"Effective Research Paper Writing: Strategies for Clarity, Coherence, and Impact"* on 22nd November 2024. The session was led by Dr. P. Rajesh, Associate Professor, Department of Physics, SSn College of Engineering. The seminar focused on equipping participants with practical techniques to enhance the structure, clarity, and overall effectiveness of academic writing. Dr. Rajesh shared valuable insights on organizing research content, improving language precision, and aligning manuscripts with journal expectations. The session was highly interactive and immensely beneficial for budding researchers and faculty members aiming to publish in reputed journals.

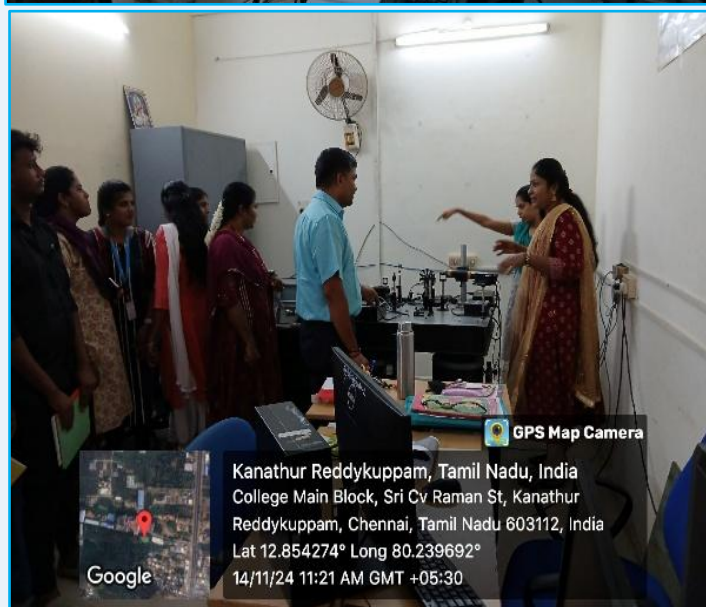
Event Organized – Seminar on Research Methodology



Seminar on Research Methodology

The Department of Physics organized a **Seminar on Research Methodology** on 15th November 2024, aimed at enhancing the research skills of students and faculty members. The session was conducted by **Dr. N. Kanagathara**, Professor, Department of Physics, Institute of Science and Humanities, Saveetha School of Engineering, Chennai. Dr. Kanagathara provided valuable insights into the fundamentals of research design, literature review, data analysis, and ethical practices in scientific research. The seminar served as a practical guide for budding researchers, helping them to approach their academic work with greater clarity, structure, and rigor.

Event Organized – Workshop on UV-Vis Spectrophotometric Techniques



Workshop on “Exploring Bandgap Analysis through UV-Vis Spectrophotometric Techniques”

The Department of Physics organized a Workshop on “Exploring Bandgap Analysis through UV-Vis Spectrophotometric Techniques” on 14th November 2024. The session featured **Dr. N. Sivakumar**, Associate Professor, Sri Sairam Engineering College, Tambaram, Chennai, as the expert speaker. The workshop focused on the principles and practical applications of UV-Vis spectroscopy for determining the optical bandgap of materials an essential technique in material science and nanotechnology research. Dr. Sivakumar shared valuable insights and hands-on approaches, making the session highly informative for students and researchers involved in optical and semiconductor material studies.

Event Organized – Onam Celebration – Onam 2K24



Onam Celebration - Onam 2K24 at AMET University

The Department of Physics organized a vibrant **Onam Celebration - Onam 2K24** at AMET University on 6th September 2024. The event was a beautiful display of Kerala's cultural heritage, with Kerala students leading the festivities and sharing their traditions. Students from various departments also enthusiastically participated, making the celebration a delightful blend of cultural diversity. The event featured traditional Onam activities like **Pookalam (flower rangoli)**, **Onam Sadhya (feast)**, and lively games, creating a festive and inclusive atmosphere. The celebrations not only honored the rich traditions of Kerala but also fostered unity among the students, strengthening the spirit of cultural exchange on campus.